Macademia nut

Actuator design:

$F_{crack} = 2179.888N$

$F_{act} = 661N$

$L_{pivot} = 45mm$

$L = 144mm$

from website uses

steel = 200GPa

a) maximum elastic deflection.

- the nutcracker is linear, length of 144mm
- force 661N applied by actuator (according to my design at the end)
- So deflection should occur where the force is applied or closer to its application
- deflection occurs at the end

b) deflection less than 2%

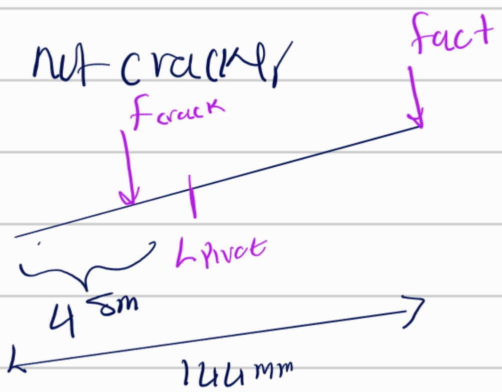
$\delta_{max} < 0.02(144mm)$

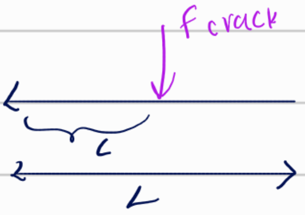
$\delta_{max} < 2.88mm$

Appendix E

$\delta_{act} = \dfrac{F_{act} L^3}{3EI}$ $\quad$ $\delta_{nut} = \dfrac{F_{crack} L^3}{3EI} (L - l)$ $\quad$ $2 \cdot 10^6 \ N/mm^2$

nut cracker





$0 < x < L$

$\Sigma M_z = 0$

$M_1 - F(L-x) = 0$

$M_1 = F(L-x)$

$$\int EI \frac{d^3y}{dx^3} = \int f(L-x)$$

$$EI \frac{dy}{dx} = f\left(Lx - \frac{x^2}{2}\right) + C1 \qquad y'(0) = 0$$

$$C_1 = 0 \Rightarrow EI \frac{dy}{dx} = f\left(ax - \frac{x^2}{2}\right)$$

$$EIy = f\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right) + C2 \qquad y = 0$$

$$C2 \Rightarrow 0$$

$$EIy = f\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right) \qquad x = L$$

$$y = \frac{f}{EI}\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right)$$

$$y(a) = \frac{f}{EI}\left(\frac{L^3}{2} - \frac{L^3}{6}\right) = \left(\frac{3L^3 - L^3}{6}\right)\frac{f}{EI}$$

$$y(a) = \frac{fL^3}{3EI}$$

$f_{crack} = 2179.888N$

$f_{act} = 661N$

$L_{pivot} = 45mm$

$L = 144mm$

$$\delta_{pivot} = \frac{f_{crack} L^3}{3EI}$$

$$\delta_{max} = \frac{f_{crack} L^3 (L-L)}{3EI} - \frac{f_{act} L^3}{3EI}$$

$$3EI\,\delta_{max} = f_{crack} L^3 - f_{act} L^3$$

$$I = \frac{f_{crack} L^3 - f_{act} L^3}{3E\,\delta_{max}}$$

$$I = \frac{661N(144mm^3) + 2179.8(144mm^3)}{3(2 \cdot 10^5 N/mm^2)(2.8mm)}$$

$I > 2624mm^4$ } too

$b = 180$

$h = 175$

$I = \frac{1}{12}bh$

you get 2625

$b = 185mm$ $\quad I = \frac{1}{12}(185 \cdot 175) = 264291 mm^4$

$h = 175 mm$

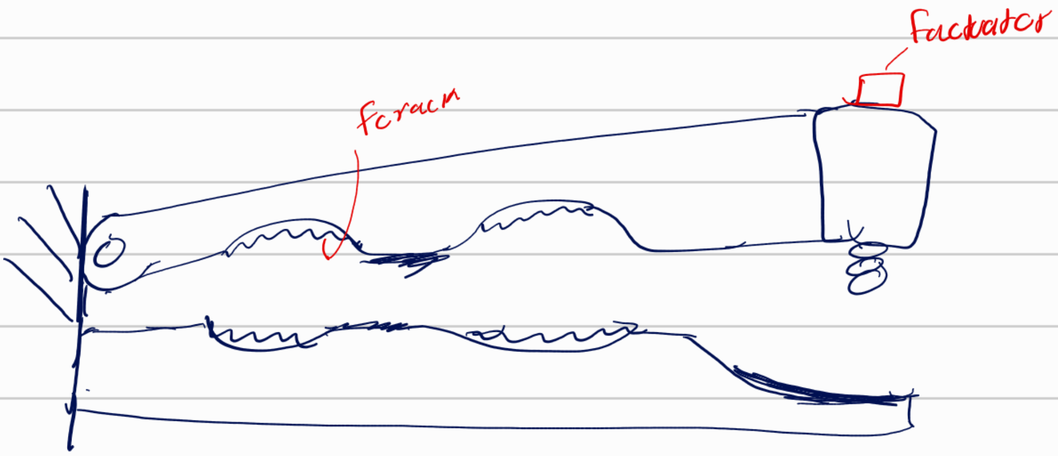
Assumption.

- the nutcracker as stated in problem to be fixed at a beam. the actuater increase the force applied to break nut, so to find max deflection the difference of nutcracker force and the force of actuator. Using my previous design length i found δ_{max} is 2.88mm
  Since we've done lot of pratice on beam cross section

  $I = 1/12\, bh$

  my $I \geq 2624 mm^4$, the b and h was check & plug till get to closest number of I and increase until inertia greater than what's found.

Nutcrack final design



175mm

185mm